

Estimations of Quantitative Features for Cuneiform Tablets

Description

This file contains the code for estimating the feature values (TTR, unigram entropy, entropy rate, repetition rate) per sign sequence on proto-cuneiform tablets from CDLI. These features were tested on a wider sample of linguistic and non-linguistic sign sequences in Bentz (2023). The repetition measure was tested also in Sproat (2014).

Session Info

Give the session info (reduced).

```
## [1] "R version 4.4.0 (2024-04-24)"  
## [1] "x86_64-pc-linux-gnu"
```

Load libraries

If the libraries are not installed yet, you need to install them using, for example, the command: `install.packages("ggplot2")`. For the Hrate package this is different, since it comes from github. The devtools library needs to be installed, and then the `install_github()` function is used.

```
library(stringr)  
library(entropy)  
library(gsubfn)
```

```
## Loading required package: proto
```

```
#library(devtools)  
#install_github("dimalik/Hrate")  
library(Hrate)
```

Give the package versions.

```
##   Hrate  gsubfn  proto entropy stringr  
##   "0.2"   "0.7"  "1.0.0" "1.3.1" "1.5.1"
```

External functions

```
# repetition rate function rrate()  
source("~/Github/PaleoSigns/code/functions/repRate_function.R")
```

Load files

Load the files with the cleaned sequences. A period needs to be chosen here. Hence, the code needs to be run for periods separately.

```
# choose period (UrukIII, UrukIV, UrukV)
period <- "UrukIII"

# cuneiform data from CDLI
cunei.data <- read.csv(paste("~/Github/PaleoSigns/data/CDLI/",
                             paste(period, "/signs_clean.csv", sep = ""),
                             sep = ""))

# head(cunei.data)
```

Cuneiform Estimations

Estimations of quantitative features for the cuneiform data from CDLI.

```
# set counter
counter = 0

# initialize dataframe to append results to
estimations.df <- data.frame(tablet.id = character(0),
                              num.tokens = numeric(0),
                              num.types = numeric(0),
                              huni.chars = numeric(0),
                              hrate.chars = numeric(0),
                              ttr.chars = numeric(0),
                              rm.chars = numeric(0))

# start time
start_time <- Sys.time()
for (i in 1:nrow(cunei.data)){
  try({ # if the processing fails for a certain row, there will be no output for this row,
    # but the try() function allows the loop to keep running

    # collect info from row
    tablet.id <- cunei.data$tablet.id[i]
    coding.sample <- as.character(cunei.data$signs.expand.vec[i])

    # split sequences per tablet into cuneiform signs
    # this is here done by using white spaces
    # note: sign transliterations consist of multiple characters,
    # e.g. NO1, N14, SZE~a, etc.
    chars <- unlist(strsplit(coding.sample, " "))

    # get number of tokens and types, i.e. "characters" are proto-cuneiform signs in this case
    num.tokens <- length(chars)
    num.types <- length(unique(chars))

    # preprocess the signs with bag-of-words normalization
    # note: this is an important step! Without it, entropy rate estimation will fail
    # to yield realistic results due to issues with processing special characters.
    chars <- PreprocessText(chars)
```

```

# unigram entropy estimation
# calculate unigram entropy for characters
chars.df <- as.data.frame(table(chars))
# print(chars.df)
huni.chars <- entropy(chars.df$Freq, method = "ML", unit = "log2")

# entropy rate estimation
# the values chosen for max.length and every.word will crucially impact processing time
# in case of "max.length = NULL" the length the total length of the sequence (n)
# use the try() function since entropy rate estimation can fail when particular
# special characters are in the strings
hrate.chars <- try(get.estimate(text = chars, every.word = 1, max.length = NULL))
if (class(hrate.chars) == "numeric") {
  hrate.chars <- hrate.chars
} else {
  hrate.chars = "NA"
}
# calculate type-token ratio (TTR)
ttr.chars <- nrow(chars.df)/sum(chars.df$Freq)

# calculate repetition rate
rm.chars <- rrate(chars, method = "lsmo")

# append results to dataframe
local.df <- data.frame(tablet.id, num.tokens, num.types, huni.chars, hrate.chars,
                        ttr.chars, rm.chars)
estimations.df <- rbind(estimations.df, local.df)
}) # end of try()
# counter
counter <- counter + 1
# print(counter)
}
end_time <- Sys.time()
end_time - start_time

```

Time difference of 4.684855 secs

Merge estimations with original file

```
cunei.data.est <- merge(cunei.data, estimations.df, by = "tablet.id")
```

Give head of estimation data frame (selected columns) for visual checking of the output data structure.

```
# head(cunei.data.est[c("tablet.id", "signs.orig.vec", "num.tokens", "num.types",
#                      "huni.chars", "hrate.chars", "ttr.chars", "rm.chars")])
```

Save outputs to file.

```
write.csv(cunei.data.est, paste("~/Github/PaleoSigns/results/features/cuneiform_",
                                paste(period, "_features.csv", sep = ""),
                                sep = ""), row.names = F)
```

Package References

```
## Bentz C, Alikaniotis D (2017). _Hrate: Implements LZ estimator for
## texts_. R package version 0.2, commit
## b67b4bb5f5430ce42d2844da30ddf5642540aa30,
## <https://github.com/dimalik/Hrate>.

## Wickham H (2023). _stringr: Simple, Consistent Wrappers for Common
## String Operations_. R package version 1.5.1,
## <https://CRAN.R-project.org/package=stringr>.

## Grothendieck G (2018). _gsubfn: Utilities for Strings and Function
## Arguments_. R package version 0.7,
## <https://CRAN.R-project.org/package=gsubfn>.

## Hausser J, Strimmer K (2021). _entropy: Estimation of Entropy, Mutual
## Information and Related Quantities_. R package version 1.3.1,
## <https://CRAN.R-project.org/package=entropy>.
```

Literature References

Bentz, C. (2023). The Zipfian Challenge: Learning the statistical fingerprint of natural languages. In Proceedings of the 27th Conference on Computational Natural Language Learning (CoNLL), pp. 27-37.

Sproat, R. (2014). A statistical comparison of written language and nonlinguistic symbol systems. *Language*, 457-481.